SEPARATION 1

Separating solids of different sizes

When a solid mixture is made up of particles of different sizes, a sieve can be used to separate the mixture.

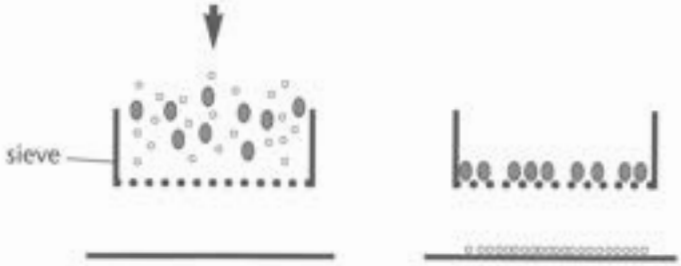


Figure 1

1 Figure 1 shows two stages in the process of using a sieve to separate a solid mixture. Explain the principle behind this method of separation.

2 Give an example of a mixture that could be separated using this method.

Separating an insoluble solid from a liquid

Filter paper is porous; it has tiny holes in it. The holes (pores) allow small particles to pass through, but hold back larger particles. Water and solutions will pass through filter paper but undissolved particles will not.

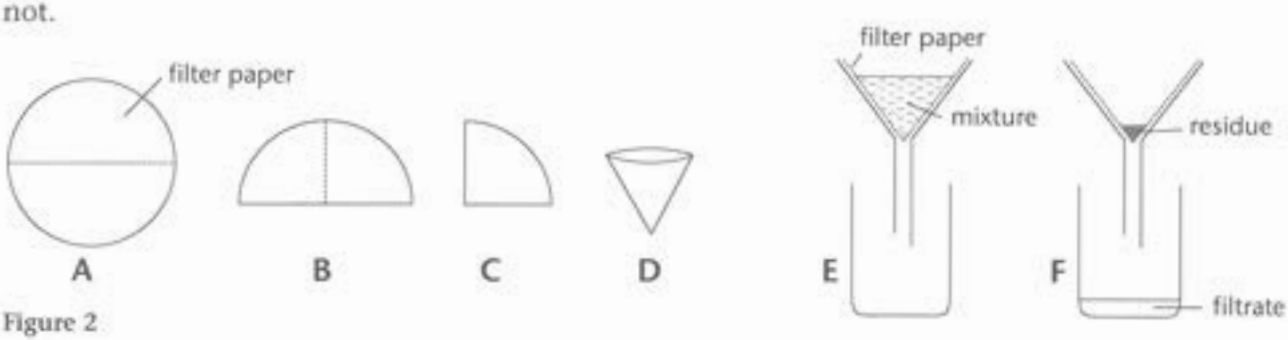


Figure 2

3 Diagrams A to F in figure 2 show a sequence for filtering a mixture of water and an insoluble solid. Refer to these diagrams to help you describe how a typical filtration could be conducted.

4 Explain what is meant by 'residue' and 'filtrate' as labelled in diagram F.

Separating dissolved solids from liquids

A solid can be recovered from a solution through evaporation (vaporising the liquid). As the mixture is heated, the solvent (liquid) changes to a vapour and evaporates. This leaves the solute (solid).

In a science laboratory, this process is sometimes done in two stages. Direct heat is used to concentrate the solution and then a water bath is used to complete the evaporation. A water bath consists of water being boiled in a beaker. As the water boils, the steam (at 100°C) warms the evaporating basin and its contents. A water bath provides a lower temperature than direct heat from a Bunsen burner.

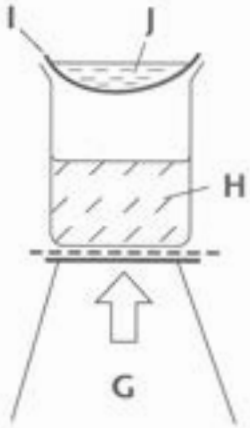


Figure 3

5 Figure 3 shows a water bath being used to evaporate a solution. Identify parts G, H, I and J.

6 Why does using a water bath to evaporate a solution take longer than using direct heat from a Bunsen burner?

7 Evaporating basins can be made of glass or ceramic materials. Give two desirable properties of a material used to make an evaporating basin.

8 Define *solute*.

9 Define *vapour*.

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Separating a soluble solid from an insoluble solid

10 The following instructions relate to separating a mixture of two solids. One solid is soluble and the other solid is insoluble. The instructions are out of sequence. Rewrite them in the correct sequence.

- **a** Heat and stir if the soluble substance is not completely dissolved.
- **b** Evaporate the filtrate to recover the soluble solid.
- **c** Add a solvent and stir.
- **d** Dry the residue on filter paper to recover the insoluble solid.
- **e** Filter and then place the filtrate in an evaporating basin.
- **f** Put the mixture into a beaker.

11 A student was given a mixture of common salt (sodium chloride) and sand. She separated these two solids using the method outlined in Question 1. Her results are shown below.

Mass of mixture:	8.3 g
Volume of water added:	50 mL
Mass of dry filter paper:	0.1 g
Mass of filter paper plus sand (after drying):	4.9 g
Mass of evaporating basin (empty):	121.6 g
Mass of evaporating basin plus salt:	125.1 g

- **a** Which solid was soluble in water?
- **b** Which solid was insoluble in water?
- **c** What was the mass of sand in the mixture?
- **d** What was the mass of salt in the mixture?
- **e** Calculate the relative percentages of sand and salt in the mixture.
- **f** Not all of the information in the results was necessary for calculating the amounts of sand and salt present. Name one piece of data that was unnecessary.
- **g** What property of filter paper allows it to separate an insoluble solid from a liquid?

The centrifuge

A **centrifuge** is a device that can separate suspensions. Centrifuges spin at high speeds (up to hundreds of thousands of revolutions per minute – rpm). Forces 10 000 times that of gravity can be obtained in a centrifuge.

A domestic spin drier is a type of centrifuge. When the machine spins, excess water is flung to the outside, off the wet clothes. At the same time warm air blows in to evaporate the water.

A **suspension** is a liquid that has small particles (liquid or solid) suspended in it. For separation, samples of a suspension are placed in centrifuge tubes. These tubes resemble test tubes but can withstand much greater forces. When the tubes are spun, the more dense particles move to the bottom of the Figure 4 shows a 'before and after' version of a mixture that has been put through a centrifuge. Centrifuges are especially important in separating biochemical mixtures such as blood.



Figure 4

12 Define *suspension*.

13 What is meant by *rpm*?

14 Outline how a centrifuge can separate a suspension.

15 Name a household appliance that uses a centrifuge.

16 Explain what is meant by *more dense*.

17 Is a blender a type of centrifuge?

Research **18** Explain why a centrifuge is important in blood preservation.

19 Find out the components in blood that can be separated using a centrifuge.

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